

Name: Fetalino Jayvee M.	Date Performed: 5/03/2026
Course/Section: CPE31S1	Date Submitted: 5/03/2026
Instructor: Engr Robin Valenzuela	Semester and SY: 2025-2026

Activity 8: Install, Configure, and Manage Availability Monitoring tools

1. Objectives

Create and design a workflow that installs, configure and manage enterprise monitoring tools using Ansible as an Infrastructure as Code (IaC) tool.

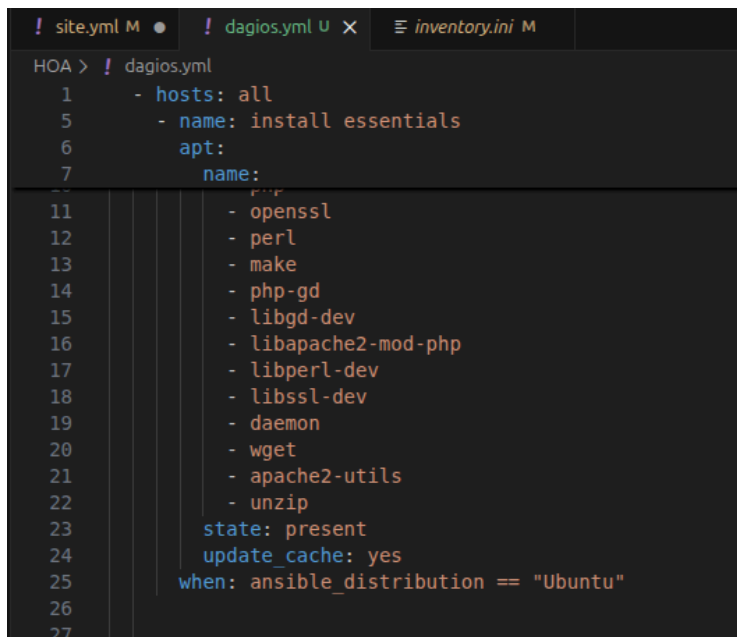
2. Discussion

Availability monitoring is a type of monitoring tool that we use if the certain workload is up or reachable on our end. Site downtime can lead to loss of revenue, reputational damage and severe distress. Availability monitoring prevents adverse situations by checking the uptime of infrastructure components such as servers and apps and notifying the webmaster of problems before they impact on business.

3. Tasks

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Nagios for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity.

4. Output (screenshots and explanations)

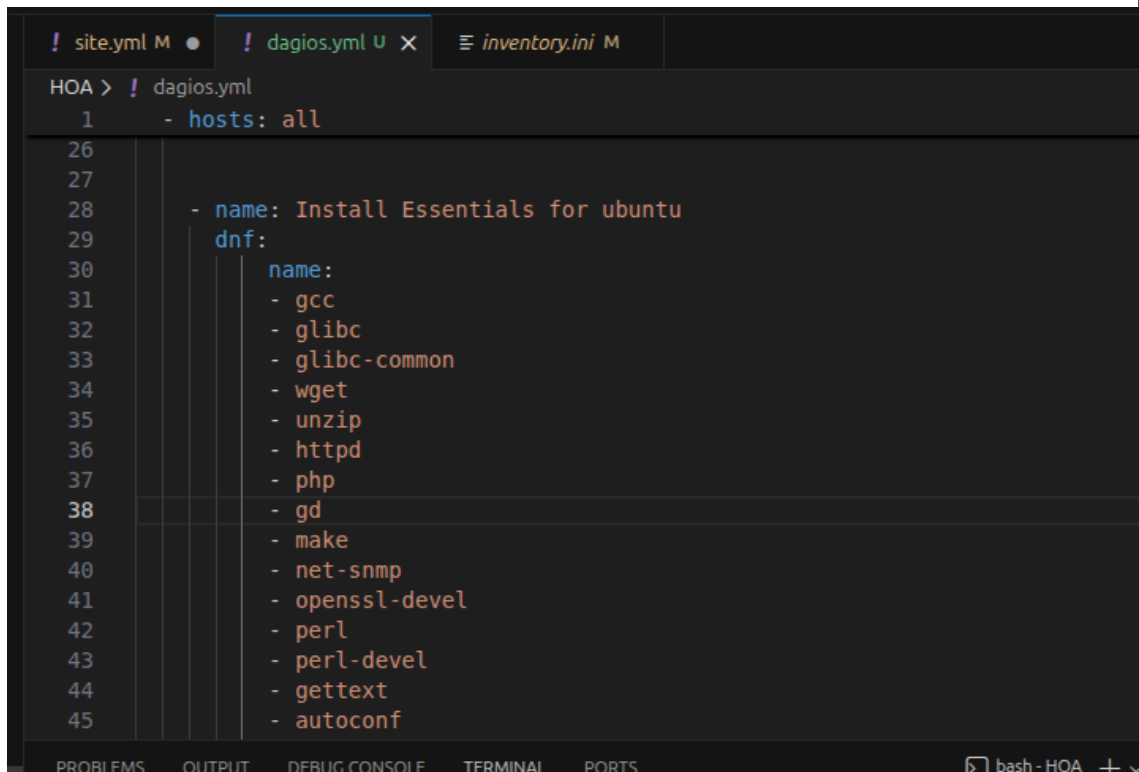


```

! site.yml M ● ! dagios.yml U x inventory.ini M
HOA > ! dagios.yml
1  - hosts: all
5    - name: install essentials
6      apt:
7        name:
11         - openssl
12         - perl
13         - make
14         - php-gd
15         - libgd-dev
16         - libapache2-mod-php
17         - libperl-dev
18         - libssl-dev
19         - daemon
20         - wget
21         - apache2-utils
22         - unzip
23       state: present
24       update_cache: yes
25       when: ansible_distribution == "Ubuntu"
26
27

```

Here I tried installing the essentials in ubuntu

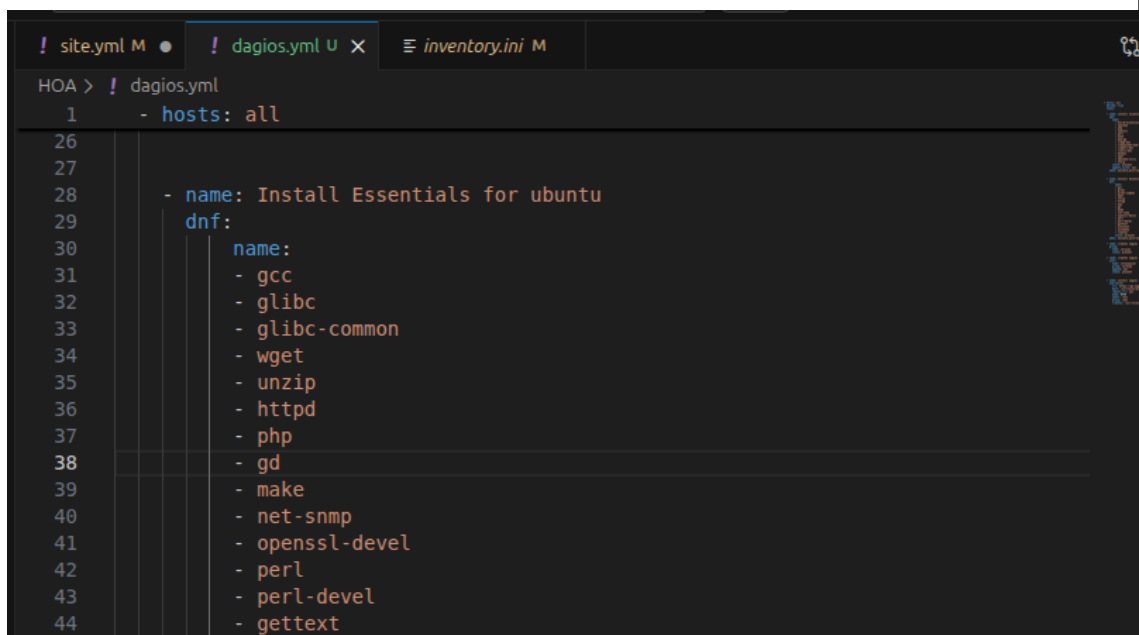


The screenshot shows a terminal window with a dark background. At the top, there are three tabs: 'site.yml M', 'dagios.yml U X', and 'inventory.ini M'. The prompt is 'HOA > ! dagios.yml'. The content of the file is as follows:

```
1 - hosts: all
26
27
28   - name: Install Essentials for ubuntu
29     dnf:
30       name:
31         - gcc
32         - glibc
33         - glibc-common
34         - wget
35         - unzip
36         - httpd
37         - php
38         - gd
39       - make
40       - net-snmp
41       - openssl-devel
42       - perl
43       - perl-devel
44       - gettext
45       - autoconf
```

At the bottom of the terminal, there is a status bar with 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. On the right side, there is a small icon and the text 'bash - HOA + v'.

Then in centos

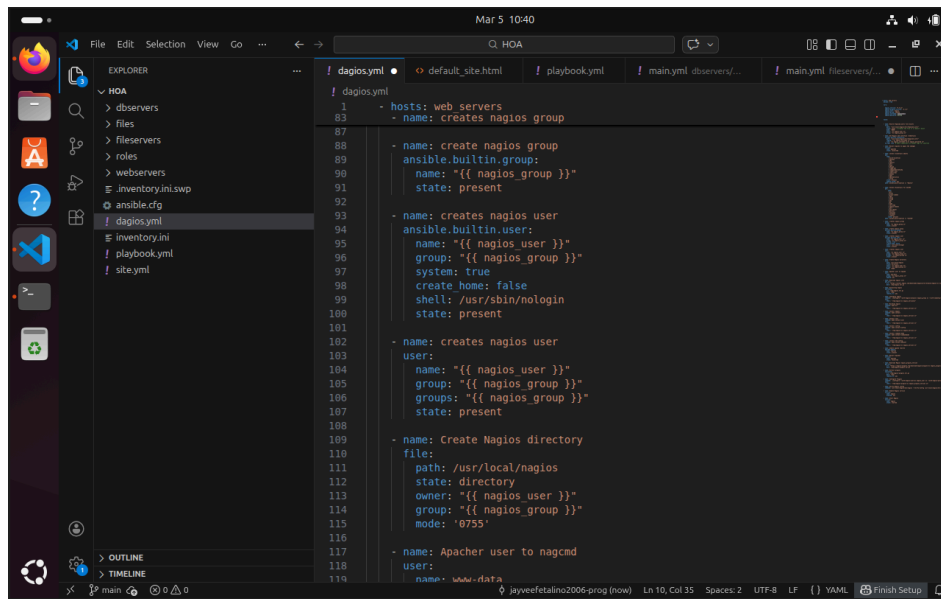


The screenshot shows a terminal window with a dark background. At the top, there are three tabs: 'site.yml M', 'dagios.yml U X', and 'inventory.ini M'. The prompt is 'HOA > ! dagios.yml'. The content of the file is as follows:

```
1 - hosts: all
26
27
28   - name: Install Essentials for ubuntu
29     dnf:
30       name:
31         - gcc
32         - glibc
33         - glibc-common
34         - wget
35         - unzip
36         - httpd
37         - php
38         - gd
39       - make
40       - net-snmp
41       - openssl-devel
42       - perl
43       - perl-devel
44       - gettext
```

At the bottom of the terminal, there is a status bar with 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL', and 'PORTS'. On the right side, there is a small icon and the text 'bash - HOA + v'.

shows system setup for Nagios by creating the nagios user and group, creating the Nagios directory with correct permissions, and allowing the web server user to use the command group so the web interface can send Nagios commands.

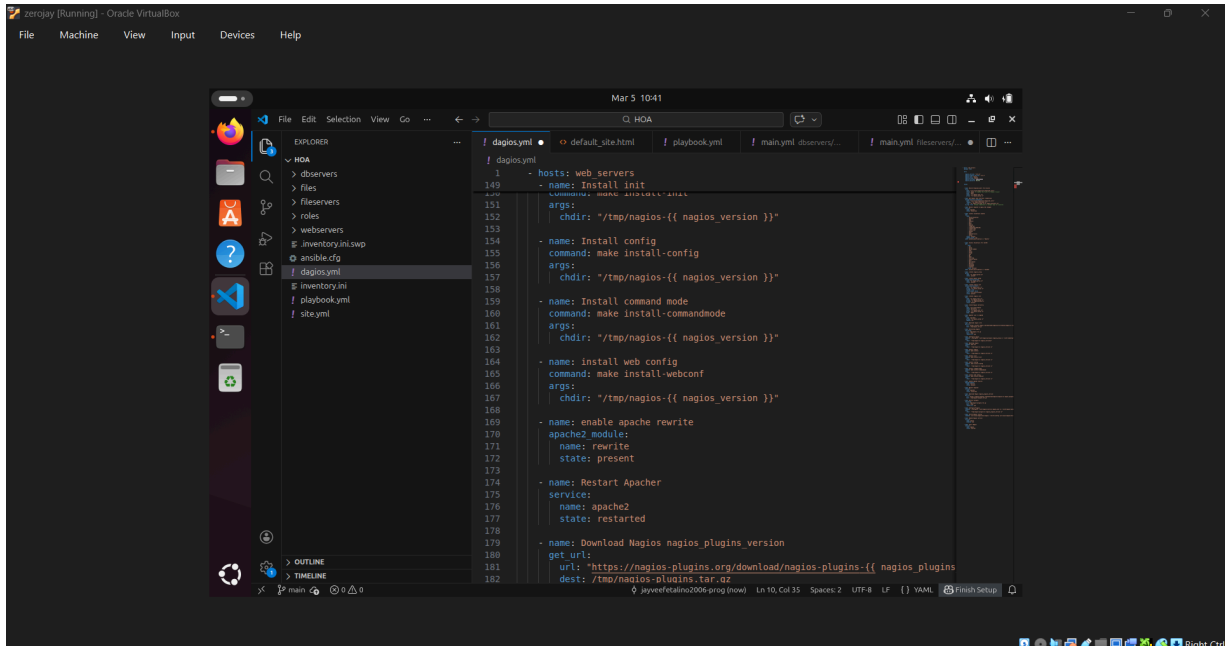


```
1  - hosts: web servers
83  - name: creates nagios group
87
88  - name: create nagios group
89    ansible.builtin.group:
90      name: "{{ nagios_group }}"
91      state: present
92
93  - name: creates nagios user
94    ansible.builtin.user:
95      name: "{{ nagios_user }}"
96      group: "{{ nagios_group }}"
97      system: true
98      create_home: false
99      shell: /usr/sbin/nologin
100     state: present
101
102  - name: creates nagios user
103     user:
104       name: "{{ nagios_user }}"
105       group: "{{ nagios_group }}"
106       groups: "{{ nagios_group }}"
107       state: present
108
109  - name: Create Nagios directory
110     file:
111       path: /usr/local/nagios
112       state: directory
113       owner: "{{ nagios_user }}"
114       group: "{{ nagios_group }}"
115       mode: '0755'
116
117  - name: Apache user to nagcmd
118     user:
119       name: www-data
```

Installing Nagios plugins by downloading, extracting, configuring, and building them, then checking the Nagios configuration and enabling and starting the Nagios service.

```
1  - hosts: web servers
109 - name: Create Nagios directory
116
117 - name: Apache user to nagcmd
118   user:
119     name: www-data
120     group: "{{ nagios_group }}"
121     append: yes
122
123 - name: Download nagios core
124   get_url:
125     url: "https://assets.nagios.com/downloads/nagioscore/releases/nagios-{{ na
126     dest: /tmp/nagios.tar.gz
127
128 - name: Extracting nagios
129   unarchive:
130     src: /tmp/nagios.tar.gz
131     dest: /tmp/
132     remote_src: yes
133
134 - name: Configure nagios
135   command: ./configure --with-nagios-group={{ nagios_group }} --with-comand-gr
136   args:
137     chdir: "/tmp/nagios-{{ nagios_version }}"
138
139 - name: Building nagios
140   command: make all
141   args:
142     chdir: "/tmp/nagios-{{ nagios_version }}"
143
144 - name: Install nagios
145   command: make install
146   args:
147     chdir: "/tmp/nagios-{{ nagios_version }}"
148
```

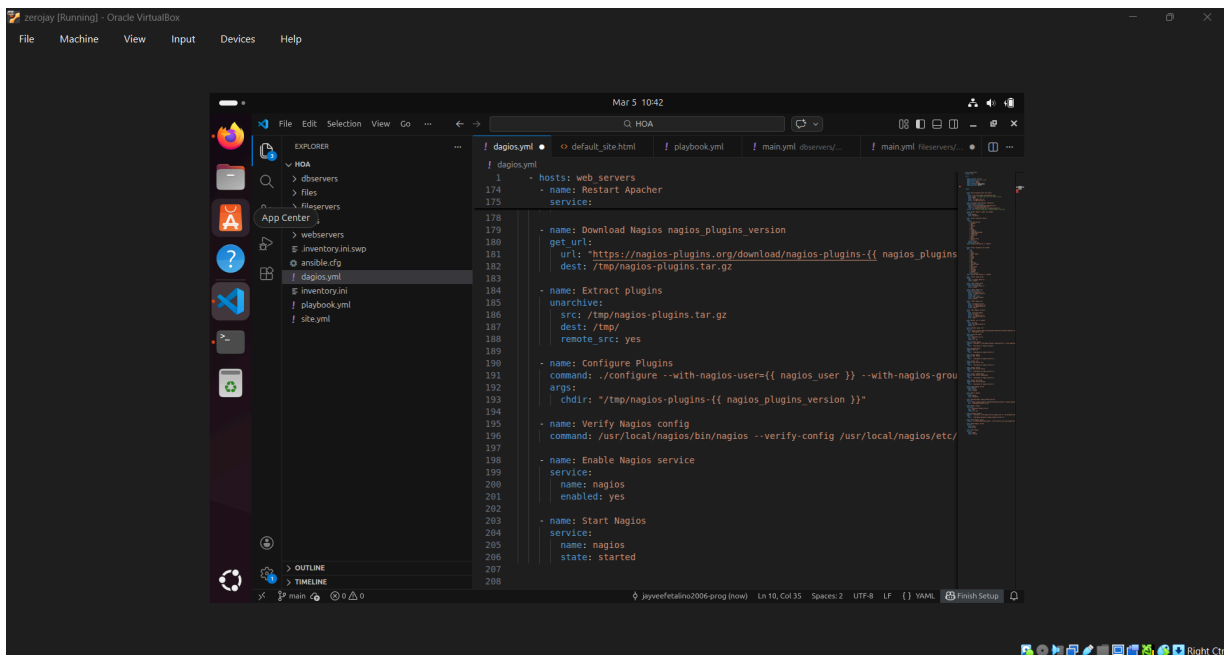
dnf prerequisites list, which is still the dependency installation needed for compiling Nagios and running its web interface.



The screenshot shows a terminal window titled "zerojay [Running] - Oracle VirtualBox" with a menu bar (File, Machine, View, Input, Devices, Help). The terminal is running a playbook named "dagios.yml" for the host "web servers". The playbook tasks are as follows:

```
1  - hosts: web servers
149 - name: Install init
150   command: make install-init
151   args:
152     chdir: "/tmp/nagios-{{ nagios_version }}"
153
154 - name: Install config
155   command: make install-config
156   args:
157     chdir: "/tmp/nagios-{{ nagios_version }}"
158
159 - name: Install command mode
160   command: make install-commandmode
161   args:
162     chdir: "/tmp/nagios-{{ nagios_version }}"
163
164 - name: install web config
165   command: make install-webconf
166   args:
167     chdir: "/tmp/nagios-{{ nagios version }}"
168
169 - name: enable apache rewrite
170   apache2_module:
171     name: rewrite
172     state: present
173
174 - name: Restart Apacher
175   service:
176     name: apache2
177     state: restarted
178
179 - name: Download Nagios nagios plugins version
180   get_url:
181     url: "https://nagios-plugins.org/download/nagios-plugins-{{ nagios_plugins
182     dest: /tmp/nagios-plugins.tar.gz
183     jayvefetalnu2006-prog (now)  Ln 10, Col 35  Spaces 2  UTF-8  LF  {}  YAML  Finish Setup
```

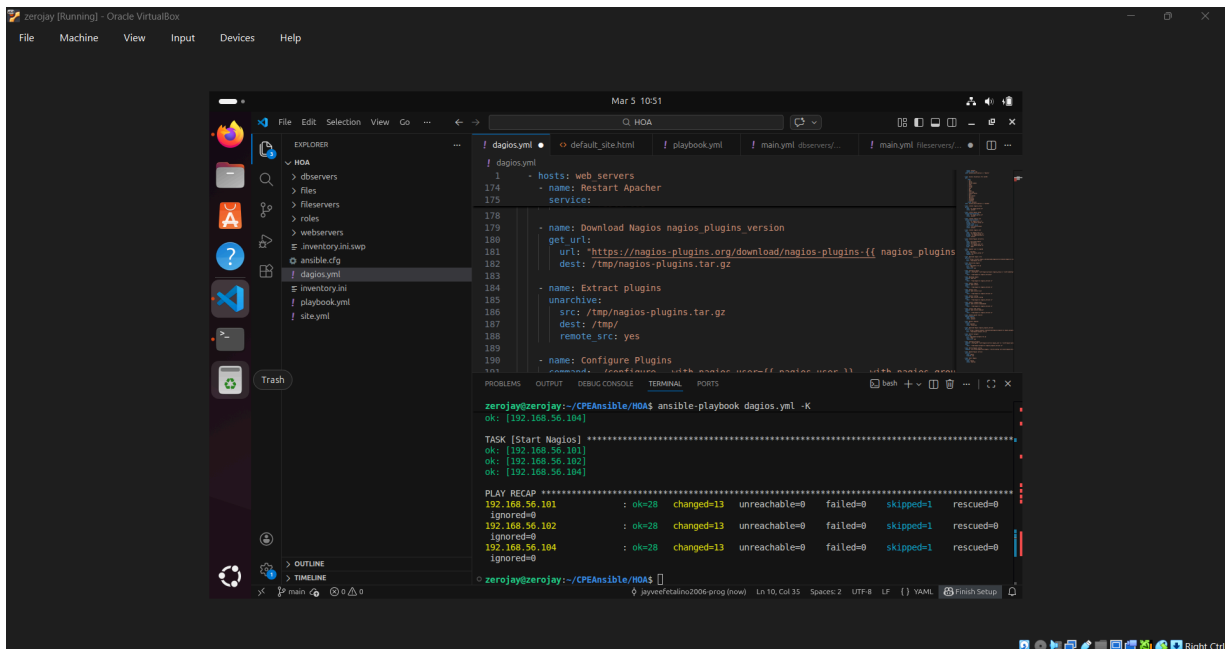
Last one shows enabling nagios service and starting it



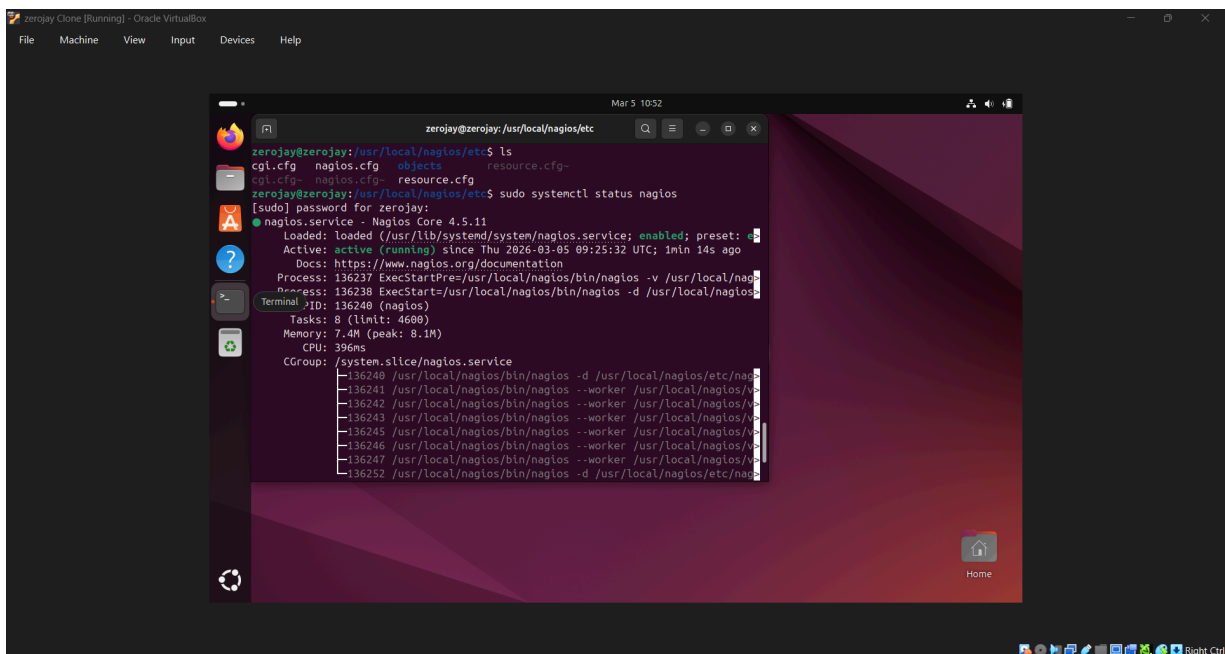
The screenshot shows the same terminal window, now at line 179 of the "dagios.yml" playbook. The tasks are:

```
179 - name: Download Nagios nagios plugins version
180   get_url:
181     url: "https://nagios-plugins.org/download/nagios-plugins-{{ nagios_plugins
182     dest: /tmp/nagios-plugins.tar.gz
183
184 - name: Extract plugins
185   unarchive:
186     src: /tmp/nagios-plugins.tar.gz
187     dest: /tmp/
188     remote_src: yes
189
190 - name: Configure Plugins
191   command: ./configure --with-nagios-user={{ nagios_user }} --with-nagios-grou
192   args:
193     chdir: "/tmp/nagios-plugins-{{ nagios_plugins_version }}"
194
195 - name: Verify Nagios config
196   command: /usr/local/nagios/bin/nagios --verify-config /usr/local/nagios/etc/
197
198 - name: Enable Nagios service
199   service:
200     name: nagios
201     enabled: yes
202
203 - name: Start Nagios
204   service:
205     name: nagios
206     state: started
207
208
```

Here shows the recap of everything when ran.



Checking for nagios in the clone



Reflections:

Answer the following:

1. What are the benefits of having an availability monitoring tool?

An availability monitoring tool helps ensure that systems and services stay running and accessible. It detects problems or downtime early and alerts administrators so they can fix issues quickly. This helps reduce service interruptions and keeps systems reliable for users.

Conclusions:

This activity demonstrated how Nagios can be installed, configured, and managed using Ansible as an Infrastructure as Code tool. By creating a

playbook and applying roles, the installation process for both Ubuntu and CentOS systems was automated and organized. The activity showed that automation simplifies system deployment, ensures consistency across servers, and reduces manual configuration errors. Overall, using Ansible with Nagios makes monitoring infrastructure easier to manage and more reliable for maintaining system availability.